When an action plateau controls a spectral gap

A complete collection of observables with bounded total susceptibility gives a lower relaxation-gap bound. A single positive velocity plateau instead gives an upper bound, and can miss an arbitrarily slow internal mode. We prove both statements for finite reversible dynamics and specify the energy normalization needed for the Dirac comparison.

1. Operator, observable and units

Let Q generate an irreducible continuous-time Markov chain on a finite set of $n \geq 2$ states, with stationary probabilities $\pi_i > 0$ and detailed balance $\pi_i Q_{ij} = \pi_j Q_{ji}$. The generator acts on functions, with row sums zero. On $L^2(\pi)$ use $\langle f, g\rangle_\pi = \sum_i \pi_i \overline{f_i} g_i$. The centered space $\mathcal{H}_0 = \{f : \langle 1, f\rangle_\pi = 0\}$ has dimension $n-1$. All domains are the entire indicated finite-dimensional space.

Set $A = -Q|_{\mathcal{H}_0}$. It is positive self-adjoint with eigenvalues $0 < \gamma_1 \leq \cdots \leq \gamma_{n-1}$. The full relaxation gap is γ_1, in inverse-time units. For a real centered observable v with velocity units, define

$$\chi(v) = \langle v, A^{-1}v\rangle_\pi = \int_0^\infty \langle v, e^{tQ}v\rangle_\pi \, dt, \qquad H(v) = 2m\chi(v), \quad m > 0.$$

χ has units length squared per time, and H has action units. This is C019's long-observation plateau for the integrated velocity. The Green–Kubo/Poisson representation is matched to Pavliotis in B20; the finite-dimensional consequences below are derived explicitly. Time here is the chain's evolution parameter.

2. One observable: the product and its direction (C041)

Write $v = \sum_j a_j e_j$ in an orthonormal eigenbasis of A, and $\sigma_v^2 = \|v\|_\pi^2 > 0$. Then

$$H(v) = 2m \sum_j \frac{|a_j|^2}{\gamma_j}, \qquad H(v)\gamma_1 \leq 2m\sigma_v^2 \leq H(v)\gamma_{n-1}.$$

Proof. Integrate each exponential in $\langle v, e^{tQ}v\rangle_\pi = \sum_j |a_j|^2 e^{-\gamma_j t}$ and use $\sum_j |a_j|^2 = \sigma_v^2$. The left inequality gives $\gamma_1 \leq 2m\sigma_v^2/H(v)$: an upper gap bound. The normalized integral time $\tau_v = \chi(v)/\sigma_v^2$ is a weighted average of $1/\gamma_j$; in particular $\tau_v \leq 1/\gamma_1$.

For the two-state chain $Q = \lambda(\sigma_x - I)$ and $v = u(1, -1)$, $\pi = (1/2, 1/2)$, with $u, \lambda > 0$, the centered space is one-dimensional. Therefore

$$\gamma_1 = 2\lambda, \qquad H(v) = \frac{mu^2}{\lambda}, \qquad H(v)\gamma_1 = 2mu^2.$$

Here σ_x exchanges the two signs. This equality identifies the source of the two-state product: every centered mode is visible to velocity.

3. Fixed plateau with a closing full gap (C041)

Take two independent signs $s, r \in \{-1, 1\}$ with rates $\lambda > 0$ and $\epsilon > 0$, respectively, and define velocity $v(s, r) = us$, with fixed $0 < u < c$. The four-state generator is

$$Q_{\lambda,\epsilon} = \lambda(\sigma_x - I) \otimes I + I \otimes \epsilon(\sigma_x - I).$$

The uniform measure is stationary and reversible. The functions $1, s, r, sr$ form an orthonormal eigenbasis with eigenvalues of $-Q$ equal to $0, 2\lambda, 2\epsilon, 2(\lambda + \epsilon)$. Velocity overlaps only the s mode. Thus

$$H(v) = \frac{mu^2}{\lambda}, \qquad \gamma_1 = 2\min(\lambda, \epsilon) \longrightarrow 0 \quad \text{as } \epsilon \downarrow 0.$$

Every positive-ϵ member is irreducible, with the same bounded speed and the same positive plateau. At the limiting parameter the label becomes conserved. The velocity path law itself is unchanged throughout the family. The missing information is the label's relaxation: finite-rate assumptions for each member give positivity member by member, while a uniform lower gap requires additional control over the family.

4. A sufficient condition: complete observability (C042)

Choose centered velocity-unit observables $f_1, \ldots, f_r$ and suppose that, for every $g \in \mathcal{H}_0$,

$$\sum_{a=1}^{r} |\langle f_a, g\rangle_\pi|^2 \geq \alpha \|g\|_\pi^2, \qquad \alpha > 0.$$

This *frame bound* means that the collection detects every centered mode; α has velocity-squared units. Put $S = \sum_a \chi(f_a)$. Then

$$\boxed{\gamma_1 \geq \frac{\alpha}{S}.}$$

If $H(f_a) \leq B_a$, it follows that $\gamma_1 \geq 2m\alpha / \sum_a B_a$.

Proof. Set $b_j = \sum_a |\langle e_j, f_a\rangle_\pi|^2$. The frame hypothesis gives $b_j \geq \alpha$ for every unit eigenvector. Hence

$$S = \sum_j \frac{b_j}{\gamma_j} \geq \frac{\alpha}{\gamma_1}.$$

Multiplication by γ_1/S proves the result. Equivalently, the frame operator $F = \sum_a |f_a\rangle\langle f_a|$ satisfies $F \geq \alpha I$ and $S = \mathrm{Tr}(A^{-1}F)$.

For a family, uniform $\alpha \geq \alpha_0 > 0$ and $S \leq S_0 < \infty$ give the uniform bound $\gamma_1 \geq \alpha_0/S_0$. When masses vary, the action-valued version must control $\sum_a B_a/(2m)$, rather than merely $\sum_a B_a$. The frame can have full rank only if $r \geq n-1$. These requirements expose the cost of carrying the estimate to larger systems.

The exact all-observable formulation is

$$\sup_{f \in \mathcal{H}_0 \setminus \{0\}} \frac{\chi(f)}{\|f\|_\pi^2} = \|A^{-1}\| = \frac{1}{\gamma_1}.$$

The spectral expansion proves the upper bound, and a slowest eigenvector attains equality. It is the inverse-operator version of a Poincaré bound.

In the four-state example choose $f_1 = us$, $f_2 = ur$, $f_3 = usr$. They give $F = u^2 I$ on $\mathcal{H}_0$, and

$$S = u^2 \left[\frac{1}{2\lambda} + \frac{1}{2\epsilon} + \frac{1}{2(\lambda+\epsilon)}\right].$$

The complete collection detects the approaching gap closure through $\chi(f_2) \to \infty$. Each observable stays bounded in magnitude by u. Thus speed control and susceptibility control address different premises.

5. Energy normalization and the companion field problem

On full $L^2(\pi)$ define $\mathcal{E} = K(-Q)$ with a supplied action unit $K > 0$. Constants form its unique zero-energy ground space; its excitation gap is $\Delta_{\mathcal{E}} = K\gamma_1$. In the two-state model, setting $K = H(v)$ gives

$$\Delta_{\mathcal{E}} = 2mu^2.$$

The numerical substitution $u = c$ matches C039's Dirac rest-branch separation $2mc^2$. The two operators have different state spaces and spectral meanings: $\mathcal{E}$ is a nonnegative finite-state relaxation operator, while the Dirac operator has positive and negative single-particle branches. A quantum field vacuum gap additionally requires its physical Hilbert space and the continuum/infinite-volume construction.

For the companion programme, C042 supplies a candidate form of estimate: observable coverage bounded below, inverse-generator response bounded above. Transferring it requires identifying the physical generator and proving that both bounds survive the relevant limits. An algorithmic sampling chain's relaxation time is a distinct object from physical Euclidean time evolution.

6. Weak access and distinct velocities (C045)

In §3's four-state model take dimensionless $0 < \delta \le 1$ and observables $f_1 = us$, $f_2 = u\delta r$, $f_3 = u\delta sr$. In the centered orthonormal basis (s, r, sr) their frame and total response are

$$F_\delta = u^2\,\mathrm{diag}(1, \delta^2, \delta^2), \quad \alpha_\delta = u^2\delta^2, \quad S_\delta = u^2\left[\frac{1}{2\lambda} + \frac{\delta^2}{2\epsilon} + \frac{\delta^2}{2(\lambda+\epsilon)}\right].$$

Set $\epsilon = \lambda\delta^2$, keeping m, u, λ fixed. Then $S_\delta = (u^2/\lambda)[1 + \delta^2/(2(1+\delta^2))]$ stays bounded, while both α_δ and the actual gap $2\lambda\delta^2$ tend to zero. This follows by diagonal substitution; full rank at each parameter is weaker than uniform calibrated coverage.

The same mechanism survives an injective physical velocity map. For $0 < \delta \le 1/4$ set

$$v_\delta(s, r) = \frac{u}{\sqrt{2}}(s + \delta r), \qquad \epsilon = \lambda\delta^2.$$

Its four values are distinct and satisfy $|v_\delta| \le 5u/(4\sqrt{2}) < u < c$. The stationary mean is zero and

$$\chi(v_\delta) = \frac{u^2}{2}\left[\frac{1}{2\lambda} + \frac{\delta^2}{2\epsilon}\right] = \frac{u^2}{2\lambda}, \qquad H(v_\delta) = \frac{mu^2}{\lambda}, \qquad \gamma_1 = 2\lambda\delta^2 \longrightarrow 0.$$

Thus the velocity itself labels all four states and defines a four-state Markov chain, with a fixed positive plateau throughout the family. To recover the observable ur from velocity, any readout g_δ satisfying $g_\delta(v_\delta(s, r)) = ur$ needs Lipschitz constant at least $\sqrt{2}/\delta$: compare the two states with the same s and opposite r. Their input separation is $\sqrt{2}u\delta$ and output separation is $2u$. Exact state identification therefore has a diverging sensitivity requirement. If this calibrated readout is available, its susceptibility is $\chi(ur) = u^2/(2\epsilon)$ and detects the closing gap.

This is a parameter-family limit, distinct from observation time or mesh refinement. Every positive-δ model has a positive gap; a uniform gap requires a uniform premise on rates or observable response and access.

7. Independent composition supplies mixed-mode control (C046)

Let Q_i be finite irreducible reversible generators on $n_i \ge 2$ states, with gaps γ_i, stationary measures π_i and centered spaces $\mathcal{H}_{0,i}$. With independent dynamics and unchanged constituent clocks, $Q = Q_1 \otimes I + I \otimes Q_2$. The centered product space is

$$(\mathcal{H}_{0,1} \otimes 1) \;\oplus\; (1 \otimes \mathcal{H}_{0,2}) \;\oplus\; (\mathcal{H}_{0,1} \otimes \mathcal{H}_{0,2}).$$

Tensoring orthonormal eigenbases shows that $-Q$ has all sums of factor eigenvalues, including zero. The first two sectors have smallest eigenvalues γ_1, γ_2; the mixed sector has minimum $\gamma_1+\gamma_2$. Consequently

$$\gamma_{\text{prod}} = \min(\gamma_1, \gamma_2) \geq \min\left(\frac{\alpha_1}{S_1}, \frac{\alpha_2}{S_2}\right)$$

when each factor has §4's frame bound α_i and total susceptibility S_i. Lifted local observables have unchanged susceptibilities, since the other factor's constant function has norm one and eigenvalue zero. They miss the entire mixed sector of dimension $(n_1-1)(n_2-1)$, so their product frame has zero minimum eigenvalue. Independence supplies the missing spectral information. A full product frame is therefore sufficient, but local frames plus the product-generator premise already yield the displayed bound.

For masses m_i, write $S_i = \sum_a H_i(f_{ia})/(2m_i)$; uniform mass-family bounds must retain these denominators. For N identical two-state factors with flip rate λ per factor, the gap is 2λ for every N. Dividing the generator by N to fix the total flip rate instead gives $2\lambda/N$. The clock choice is part of the theorem.

The tensor-eigenbasis construction is standard product-chain theory: Levin–Peres, with contributions by Wilmer, §12.4, Lemma 12.12 and Corollary 12.13, use weighted discrete random scan. Our additive continuous generator and local-frame consequence are derived above; B22 records the precise normalization and source review.

8. Next mechanical test and reproduction

G02 identifies two sufficient routes: uniformly calibrated full coverage, or local coverage with independent-factor dynamics controlling mixed modes. The next mechanical step specifies a conservative receiver and derives its correlation and measurement scales. Interactions require a fresh estimate; the independence premise must be replaced by a proved dynamical bound. A09 retains preparation independence and coherent-action scale selection.

The written proofs are the mathematical verification route. Earlier scripts are historical artifacts under the repository's hard verification rule. The B20 companion records the Green–Kubo source match and the bounded prior-art coverage. C041 is a spectral/product-chain consequence of C019; C042 is an elementary finite-dimensional observability criterion, with no novelty claim.